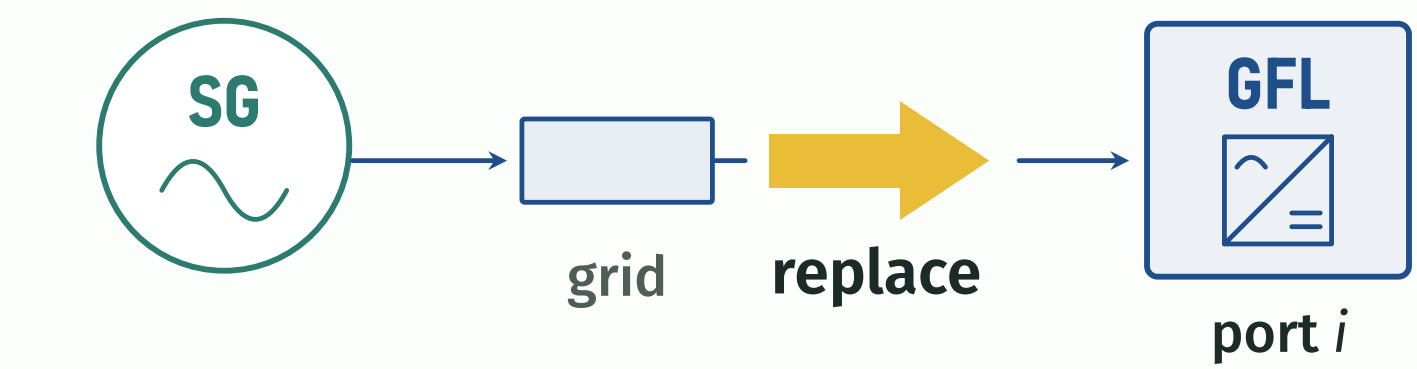


1 THE PROBLEM: SAFE SG → GFL TRANSITION

A replacement is local hardware, but stability is a property of the closed grid and its controller architecture.



$$\delta_i \in \{0, 1\}, \quad S(\delta) = \{i : \delta_i = 1\}$$

$\delta_i = 0$: retain SG; $\delta_i = 1$: replace SG; by GFL_i

Engineering objective

$$\max_{S,K} \sum_{i \in S} P_i^0 \quad \text{s.t.} \quad \alpha_{\perp}(S,K) \leq -\sigma_{\text{req}}, \quad K \in \mathcal{K}_n$$

Report the largest independently verified portfolio found.
Certified global optimality is future work.

A safe SG → GFL transition is a portfolio-controller co-design problem.

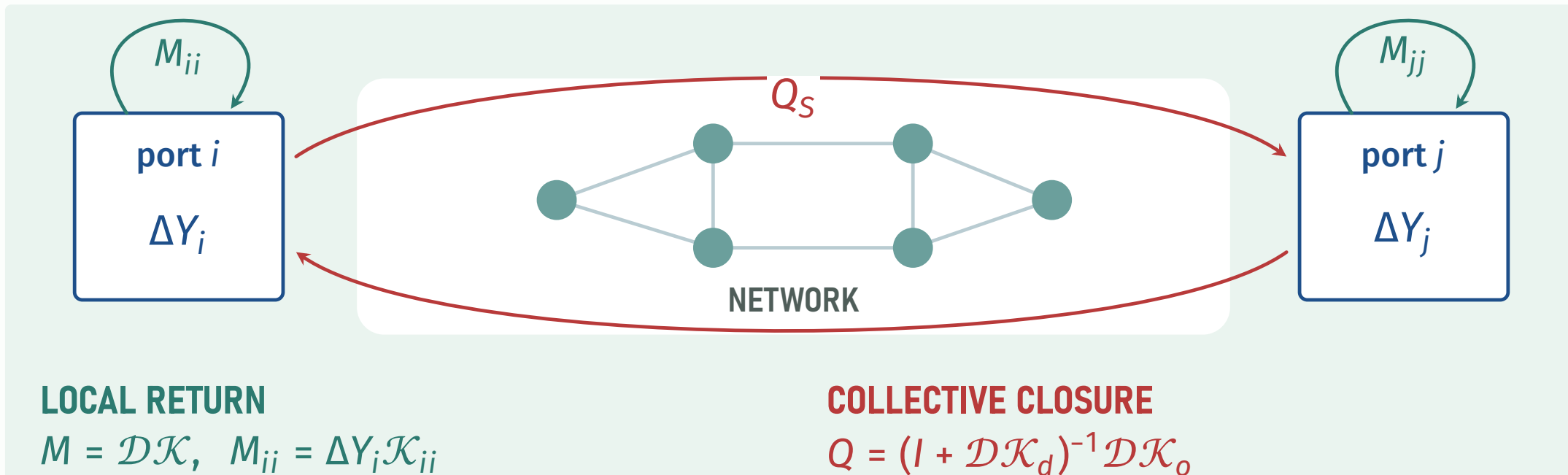
P_i^0 : dispatched power of SG_i $\alpha_{\perp}$: rightmost physical (non-gauge) root σ_{req} : required decay margin
 $\mathcal{K}_n$: PLL gains that may use measurements up to n network hops away

4 EXACT NETWORK CLOSURE: LOCAL × COLLECTIVE

$$T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$$

$$\Delta Y_i = E_i^T [T_{ij}(s) - T_0(s)] E_i, \quad \mathcal{K} = E^T T_0^{-1} E = \mathcal{K}_d + \mathcal{K}_o$$

E_i selects port i ; T_S : devices of S replaced; T_0 : all-SG network. $\mathcal{D} = \text{blkdiag}(\Delta Y_i)$: device-determined; $\mathcal{K}$: network-determined.



$$\det(I + M_S) = \prod_{i \in S} \det(I + M_{ii}) \cdot \det(I + Q_S)$$

EXACT

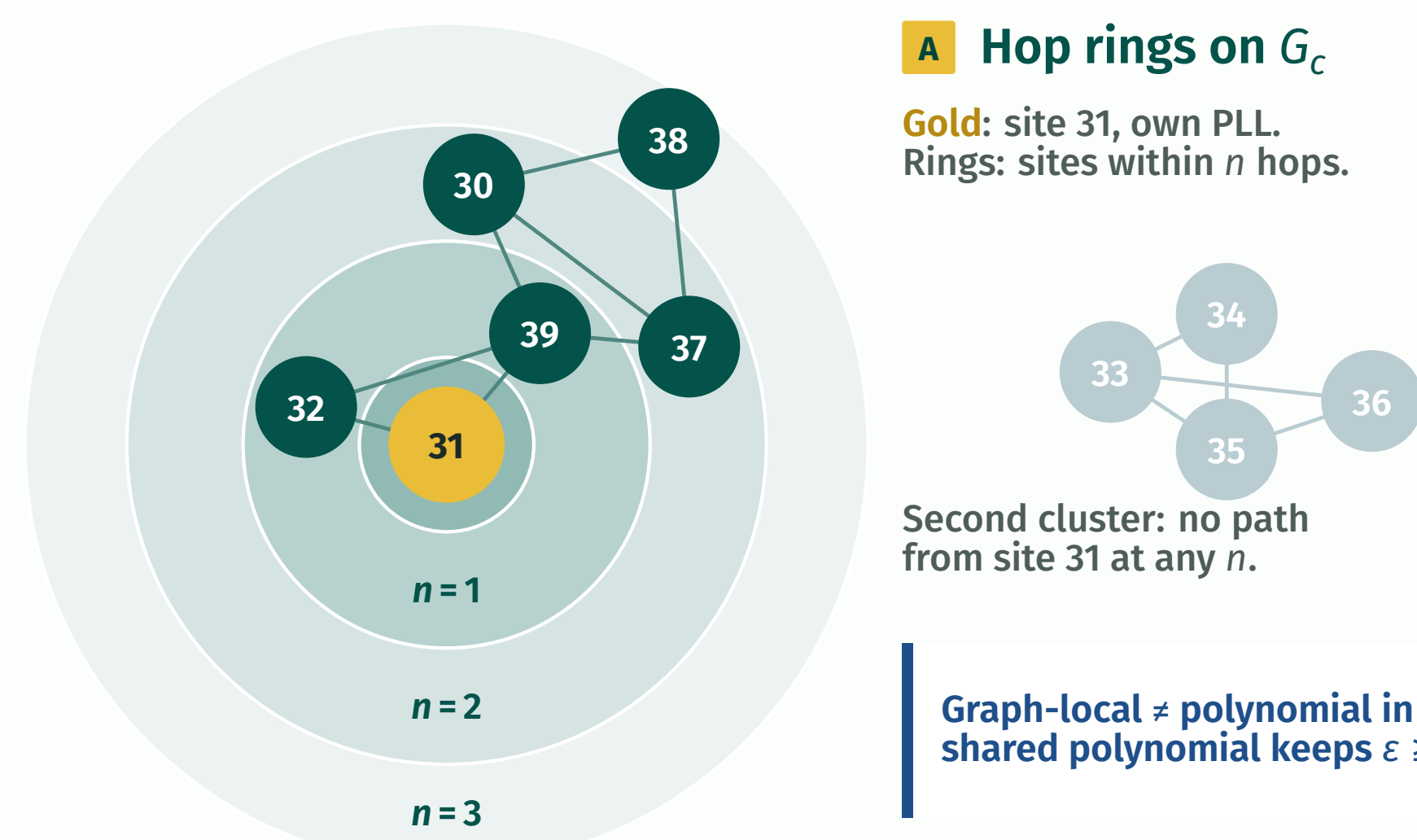
Audit: stored boundary $H_4 = \{30, 33, 35, 37\}$, $f \approx 0.7064$ Hz

$\min_i \sigma_{\min}(I + M_{ii}) = 0.4892$, $\sigma_{\min}(I + Q_{H_4}) = 4.39 \times 10^{-8}$

Delayed full IEEE-39, 44 ms: $\text{eig}(Q) = -1$ at every unstable root, with all $|L_i| \geq 0.011$.

Nothing failed locally. The network closed the destabilising loop.

7 INFORMATION RADIUS AND N-HOP CONTROL



A Hop rings on G_c
Gold: site 31, own PLL.
Rings: sites within n hops.

Second cluster: no path from site 31 at any n .

Graph-local $\neq$ polynomial in L :
shared polynomial keeps $\varepsilon \geq 9 \times 10^{-3}$.

$$\mathcal{N}_i^{(n)} = \{j : \text{dist}_{G_c}(i, j) \leq n\}$$

$$C_i^{(n)} k_i = b_i, \quad \varepsilon_i(n) = \|(I - C_i C_i^T) b_i\| / \|b_i\|$$

$$\mathcal{K}_n \subset \mathcal{K}_{n+1} \Rightarrow \varepsilon(n+1) \leq \varepsilon(n)$$

free n -hop class only

$$\hat{\xi}_i = \sum_{j \in \mathcal{N}_i^{(n)}} k_{i,j} e_j(t - \tau_j)$$

$$t_f \dot{\omega}_i = \hat{\xi}_i - \omega_i + \sum_{j \in \mathcal{N}_i^{(n)}} k_{p,i,j} e_j(t - \tau_j)$$

local $n=0$ polynomial in L free n -hop dense benchmark

MAIN CONTRIBUTIONS

- Minimal replacement blockers: the H_4 witness is unstable although all its proper subsets are stable.
- Exact closure factorization of the network return into a local and a collective part.
- Beyond nodal damping: self-energy and signed damping redistribution.

- Grid-embedded PLL gain synthesis and exact delay transport.
- Information radius on the full IEEE-39: authority grows, safety must still be verified.

2 MODEL: DAE → TRANSVERSE STABILITY

$$E(x, \delta) \dot{x} = f(x, z; \delta, K, \tau)$$

$$0 = g(x, z; \delta)$$

x dynamic states
 z network
 δ portfolio
 K gains, τ delays

$$\text{1 Operating point} \quad f(x^*, z^*) = 0, \quad g(x^*, z^*) = 0$$

$$\text{2 Linearise} \quad \begin{bmatrix} \Delta \dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} f_x & f_z \\ g_x & g_z \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta z \end{bmatrix}$$

$$\text{3 Eliminate algebra} \quad A = f_x - f_z g_z^{-1} g_x$$

Schur complement, g_z invertible

$$\text{4 Transverse quotient} \quad A_{\perp} = Z^T A Z$$

Z : orthonormal basis of $\gamma^{\perp}$ (gauge $A\gamma = 0$)

$$\alpha_{\perp}(S, K) = \max \text{Re } \sigma(A_{\perp})$$

< 0 STABLE

> 0 UNSTABLE

With PLL delay, $\alpha_{\perp}$ is the rightmost root of $\det \Delta_{\perp}(s)$;
 $\Delta(s) = sI - A_0 - B E(s) C$, exact exponentials.

PORTFOLIO > EQUILIBRIUM > DAE JACOBIAN > TRANSVERSE QUOTIENT > RIGHTMOST ROOT

5 BEYOND NODAL DAMPING: EFFECTIVE OPERATORS

A Exact Schur self-energy

EXACT

$$S_r(s) = sI - A_{rr} - A_{rc}(sI - A_{cc})^{-1} A_{cr}$$

$$T(s) = s^2 M + s D_0 + L + \Pi(s)$$

r retained, c eliminated states

B Even/odd split $\Pi = \Pi_e + \Pi_o$

PROPOSED

$$L_{\text{eff}} = L + \Pi_e(0), \quad D_{\text{eff}} = D_0 + \Pi_o(s)/s$$

$$M_{\text{eff}} = M + [\Pi_e(s) - \Pi_e(0)]/s^2$$

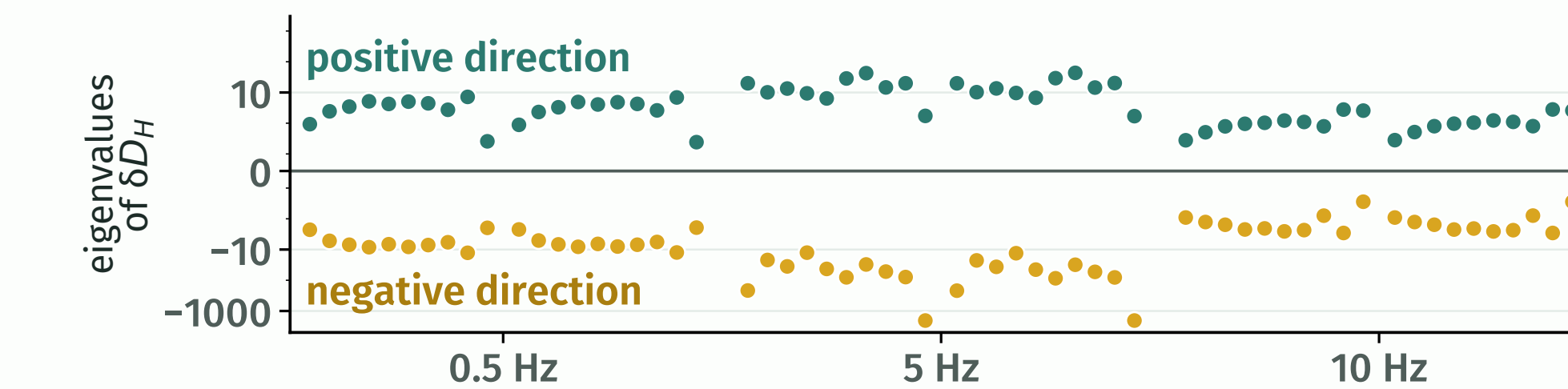
$$T = s^2 M_{\text{eff}} + s D_{\text{eff}} + L_{\text{eff}}$$

M_{eff} is an operator, not rotor inertia.

C Damping operator $D_H(\Omega) = \text{Herm } Z(j\Omega)$

$$\delta D_H = \frac{1}{2}(a v^H + v a^H)$$

one positive, one negative eigenvalue



60 cases: 10 PLL sites × 3 frequencies × 2 designs

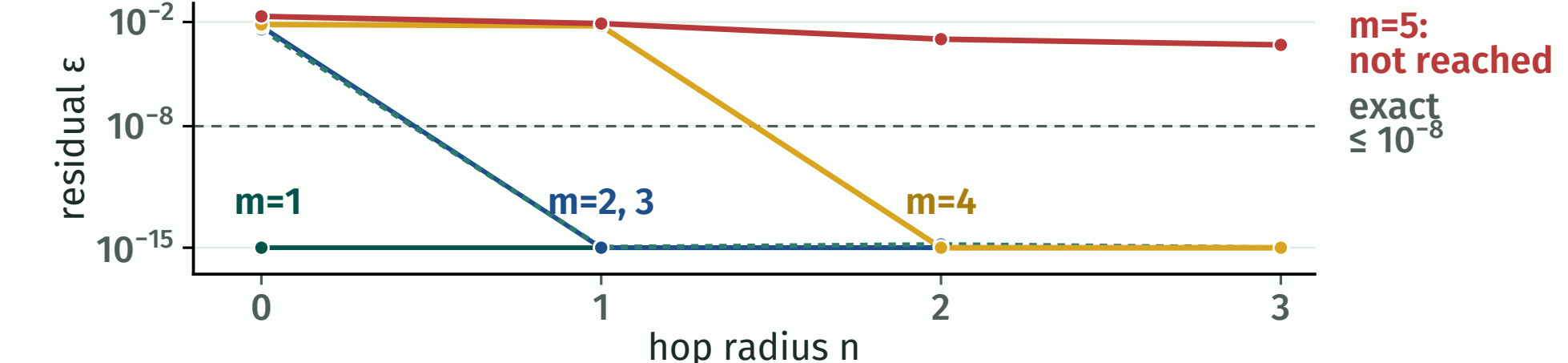
A local change moves damping between paths; damping at one frequency is no certificate.

8 N-HOP RESULTS: AUTHORITY, SPILLOVER, REPAIR

Full IEEE-39, exact delay 44 ms

NUMERICAL OBSERVATION

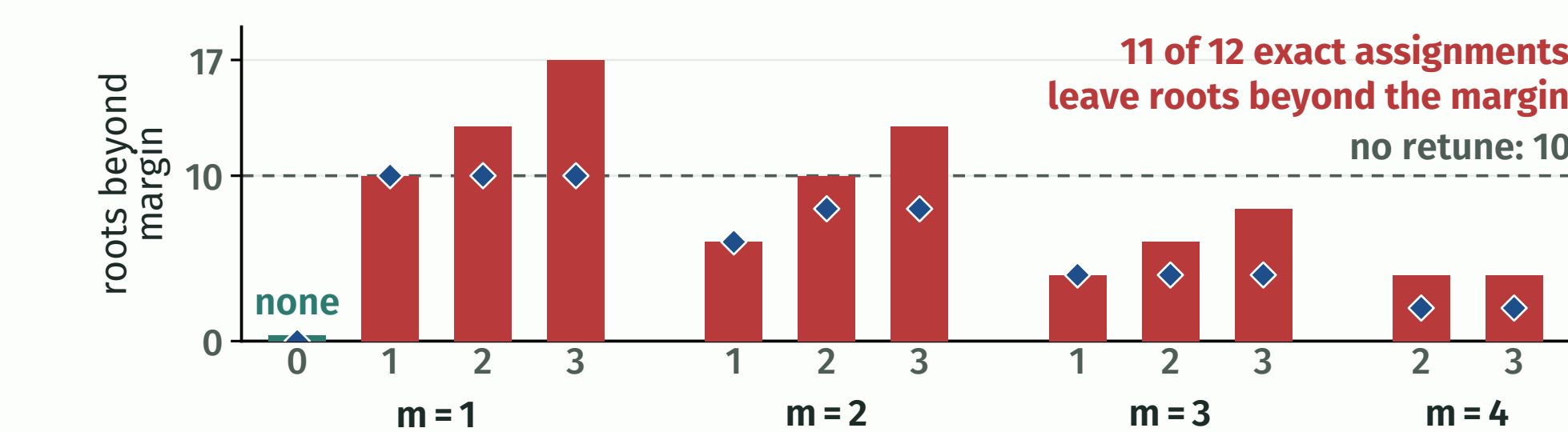
A Authority: residual vs radius



Each extra mode pinned needs more measured neighbours (dimension count $2m \leq 2|\mathcal{N}_i|$).

B Spillover on other roots

absolute gains baseline + change

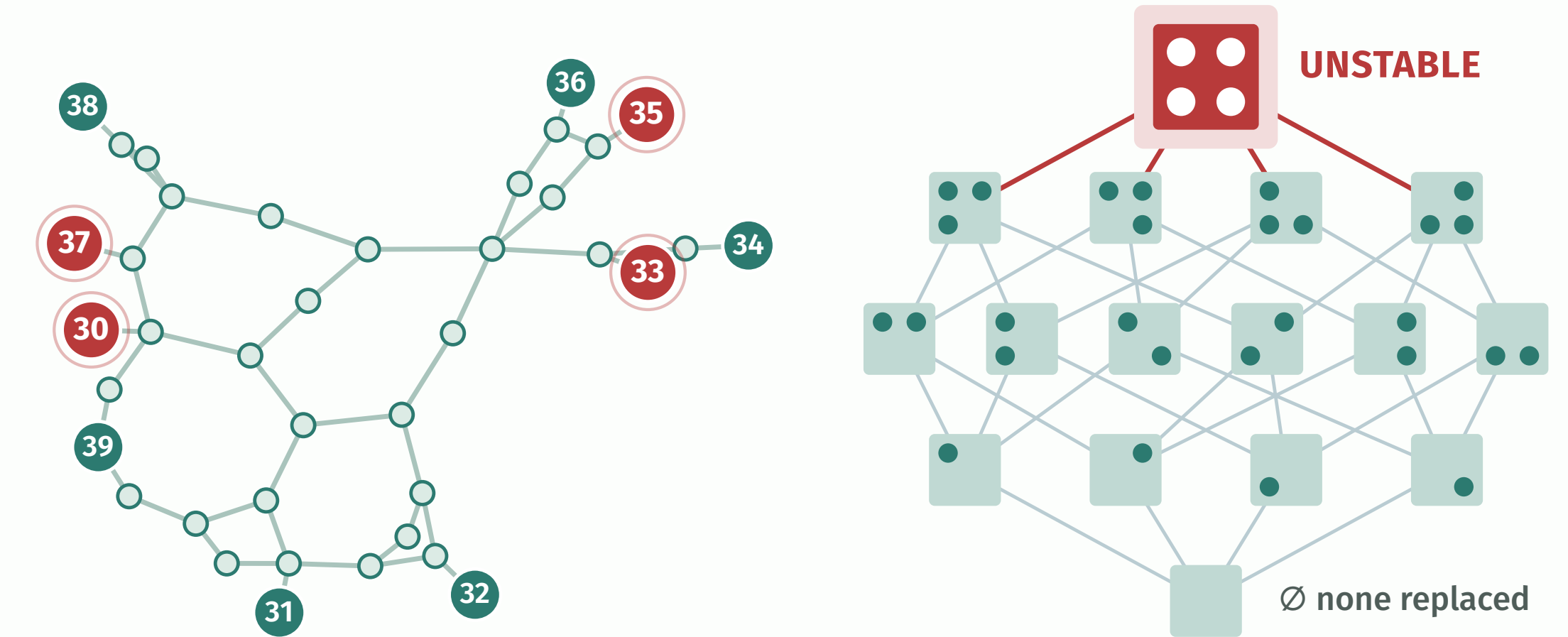


Repair at 44–52 ms already works with $n = 0$. Largest verified share: 90 % for every n ; 92.5 % fails for all n . Declared sweep, not a certified optimum.

More modal authority $\neq$ a safer grid.

3 IEEE-39: MINIMAL COLLECTIVE BLOCKER

SEPARATE TX4 FIXED-POLICY MODEL CONTRACT (NO PLL DELAY)



IEEE-39 grid. Red: the four sites of the portfolio H_4 ; teal: other SGs

Each glyph is one portfolio: filled dot = replaced site (grid 30 33 / 35 37).

Minimal collective blocker $H_4 = \{30, 33, 35, 37\}$

$$\alpha_{\perp}(H_4) = +0.127 \text{ s}^{-1} \text{ at } 0.6223 \text{ Hz}$$

15/15 proper subsets stable — full coalition unstable

Canonical nominal witness, not a universal blocker.

6 GRID-EMBEDDED PLL GAIN SYNTHESIS

$$\text{1 Open the PLL ports, target } \lambda = \alpha + j\omega$$

$$A(s; \rho)z = B(s; \rho)\theta, \quad e = Cz + D\theta \Rightarrow G = D + CA^{-1}B$$

$$\text{2 Row condition at } \lambda, \text{ pattern } q, y = G(\lambda; \rho)q$$

$$(\lambda K_{p,i} + K_{i,i}) e^{-\lambda \tau_i} y_i = \lambda^2 (1 + t_{f,i} \lambda) q_i$$

$$\text{3 } W_i = \frac{\lambda^2 (1 + t_{f,i} \lambda) e^{\lambda \tau_i} q_i}{y_i}$$

$$W_i = \lambda K_{p,i} + K_{i,i}$$

$$\text{4 } K_{p,i} = \frac{\text{Im } W_i}{\omega}, \quad K_{i,i} = \text{Re } W_i - \alpha K_{p,i}$$

$$K = \Phi(\rho, \lambda, q, \tau)$$

The gain is local; its design is embedded in the grid mode.

$$[\lambda K_p(h) + K_i(h)] e^{-\lambda(\tau+h)} = [\lambda K_p + K_i] e^{-\lambda \tau}$$

$$\left. \frac{d\mu}{dh} \right|_0 = -K_p(\mu - \lambda)(\mu - \bar{\lambda}) \frac{\partial \mu}{\partial K_i}$$

Preserving one mode does not preserve the others.

Full IEEE-39, exact delay 44 ms

NUMERICAL

no retune
8 roots
events abort

protect 4.91 Hz
0 roots
2/5 events pass

protect 1.04 Hz
0 roots
5/5 events pass

5 events per protected design; abort: integrator stopped.

9 DELAYS, CO-DESIGN, AND NONLINEAR CHECKS

A Unequal latencies mix graph modes

EXACT

$$E_{\tau}(s) = \text{diag } e^{-s\tau_i}, \quad [L, K(L)E_{\tau}] = K(L)[L, E_{\tau}]$$

$$\| [L, E_{\tau}(j\Omega)] \|_F^2 = 8 \sum_{ij \in \mathcal{E}} w_{ij}^2 \sin^2 \left(\frac{\Omega(\tau_i - \tau_j)}{2} \right)$$

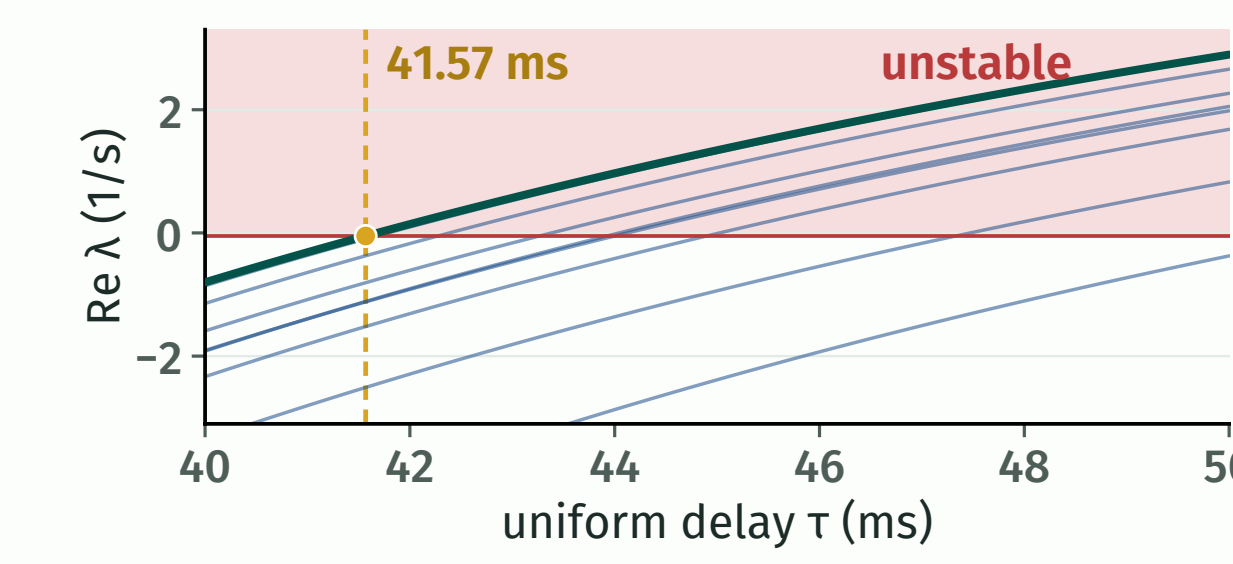
zero only if $\tau_i = \tau_j$ on every edge

C Co-design loop

- Choose the portfolio
- Equilibrium and full spectrum
- Diagnose closure Q and D_H
- Choose the radius n
- Gain map and predictor
- FULL-SPECTRUM corrector
- Nonlinear events

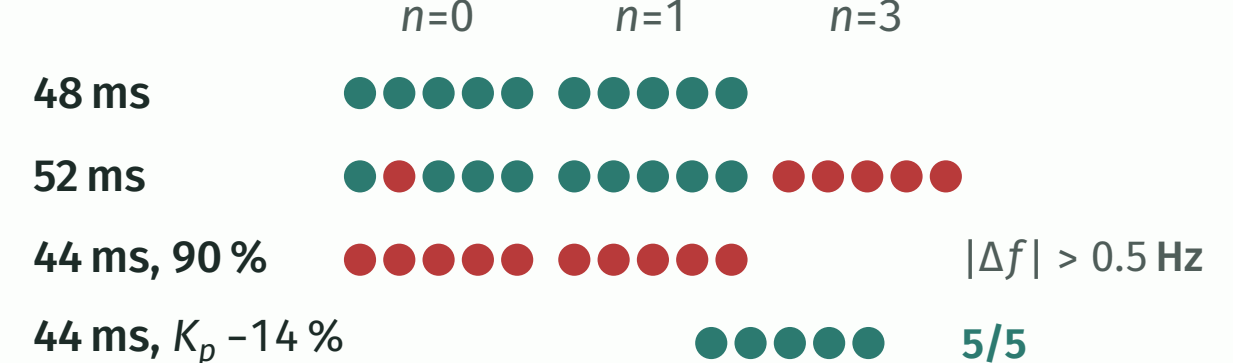
B Delay root locus

NUMERICAL



Full IEEE-39, exact delay; margin -0.05 s^{-1}

D Five frozen nonlinear events



Spectral margin is necessary, not sufficient.

SCOPE OF THE RESULTS

EXACT

closure factorization
gain map W_i
delay transport
commutator identity
signed δD_H

NUMERICAL

H_4 witness (TX4 contract)
full IEEE-39 delay campaign
 n -hop authority, spillover,
repair
nonlinear events

PROPOSED

$M_{\text{eff}}, D_{\text{eff}}$ graph
diagnostics
certified global
maximum share
EMT and current limits

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PRESERVING ONE MODE DOES NOT PROTECT THE GRID.

More information gives more control authority — only the full grid decides whether a larger SG → GFL transition is safe.